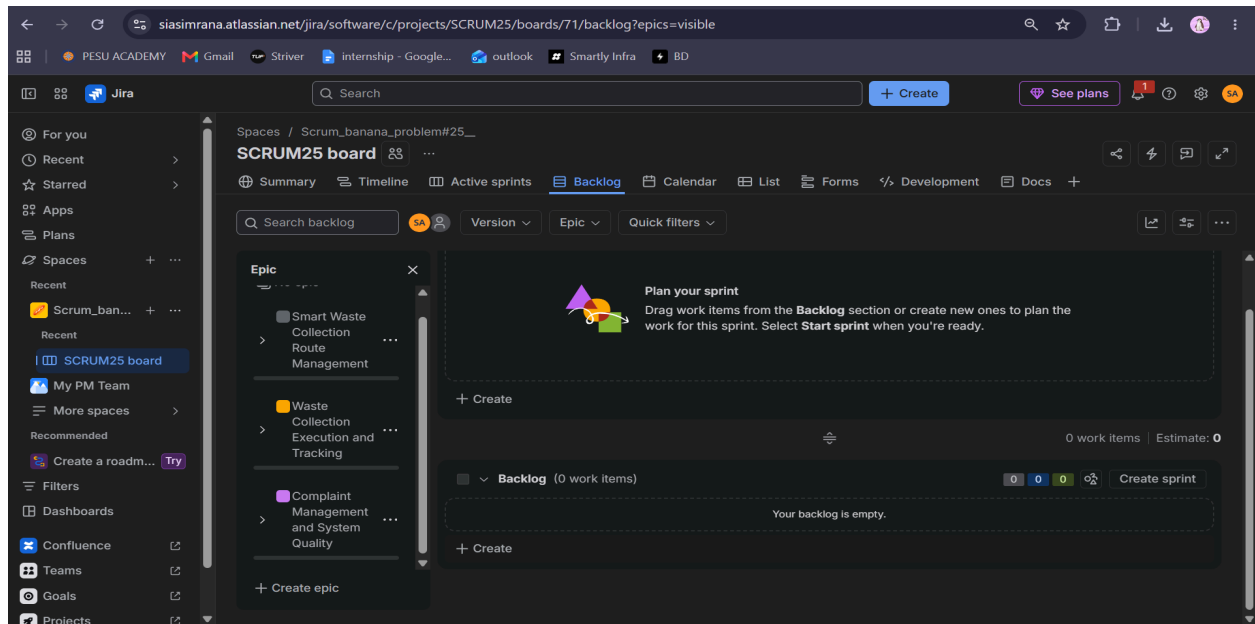


# SIA SIMRAN A\_PES1UG24CS453\_BPS#25

## 1. EPICS AND BOARD CREATION



## 2. STORY 1 SUBTASK CREATION

Jira work item

SCRUM25-1 / SCRUM25-4

1

## Generate optimized collection routes

To Do

Improve Story

Description

As a Sanitation Supervisor,  
I want the system to generate optimized daily collection routes for bins that have exceeded the 80% fill threshold,  
So that overflowing bins can be prioritized and collected efficiently.

Subtasks

Create suggested work items

Suggest

0% Done

| Work                                                              | Pri... | Stor... | Ass...                   | Status |
|-------------------------------------------------------------------|--------|---------|--------------------------|--------|
| <div><div></div>SCRUM25-11</div> Retrieve smart bin fill-level... | =      | M.      | <div><div></div>Ui</div> | To Do  |
| <div><div></div>SCRUM25-12</div> Filter bins above 80% f...       | =      | M.      | <div><div></div>Ui</div> | To Do  |

Name this sub-task

Sub-task

Cancel

Linked work items

Add linked work item

### 3. STORY 2 SUB TASK CREATION

☒ Jira work item

SCRUM25-1 / ☒ SCRUM25-5

1

## Manually modify collection routes

To Do ▾

⚡

⬆ Improve Story

⋮

+

▼ Description

As a Sanitation Supervisor,

I want to manually add or remove bins from a generated collection route,

So that I can handle special situations and adjust routes when necessary.

▼ Subtasks

⋮ ▢ +

🔗 Create suggested work items

Suggest

0% Done

| Work                                               | Pri... | Stor... | Ass... | Status  |
|----------------------------------------------------|--------|---------|--------|---------|
| <a href="#">SCRUM25-13</a> Add bin to generated... | = M.   |         | UI     | To Do ▾ |
| <a href="#">SCRUM25-14</a> Remove bin from...      | = M.   |         | UI     | To Do ▾ |

🔗 Sub-task ▾

↩

Cancel

Linked work items

Add linked work item

#### 4. STORY 3 SUB TASK

☒ Jira work item

SCRUM25-2 / ☒ SCRUM25-6

1

## View assigned collection route

To Do ▾

⚡

✦ Improve Story

⋮

+

▼ Description

As a Truck Driver,  
I want to view my assigned daily collection route and list of bins,  
So that I know which locations need to be visited for waste collection.

▼ Subtasks

⋮ ▢ +

✦ Create suggested work items

Suggest

0% Done

| Work                                                     | Pri... | Stor... | Ass... | Status  |
|----------------------------------------------------------|--------|---------|--------|---------|
| <a href="#">SCRUM25-15</a> Display assigned...           | = M.   |         | UI     | To Do ▾ |
| <a href="#">SCRUM25-16</a> Display list of assigned bins | = M.   |         | UI     | To Do ▾ |

Name this sub-task

[Sub-task](#) ▾

↵

Cancel

### Linked work items

Add linked work item

Confluence content ⓘ

⋮ +

5. STORY 4

☒ Jira work item

⚡ SCRUM25-3 / ☒ SCRUM25-8

1

## Manage missed-pickup complaints

To Do ▾

⚡

⚡ Improve Story

⋮

+

▽ Description

As a Sanitation Supervisor,

I want to view and update the status of missed-pickup citizen complaints,

So that missed waste collections can be tracked and resolved.

▽ Subtasks

⋮ +

⚡ Create suggested work items

Suggest

0% Done

| Work                                                   | Pri... | Stor... | Ass... | Status  |
|--------------------------------------------------------|--------|---------|--------|---------|
| <a href="#">SCRUM25-17</a> Create mark-as-collected... | = M.   |         | UI     | To Do ▾ |
| <a href="#">SCRUM25-18</a> Update bin collection...    | = M.   |         | UI     | To Do ▾ |

Sub-task ▾

Cancel

### Linked work items

Add linked work item

Confluence content

⋮ +

## 6. STORY 5

☒ Jira work item

SCRUM25-3 / ☒ SCRUM25-10

1

### Implement role-based system access

To Do ▾

⚡

✦ Improve Story

...

+

▼ Description

As a system administrator,

I want users to access only the features authorized for their assigned role,

So that the system remains secure and unauthorized access is prevented.

▼ Subtasks

... ▢ +

✦ Create suggested work items

Suggest

0% Done

| Work                                           | Pri... | Stor... | Ass... | Status  |
|------------------------------------------------|--------|---------|--------|---------|
| <a href="#">SCRUM25-19</a> Display missed-...  | = M.   |         | UI     | To Do ▾ |
| <a href="#">SCRUM25-20</a> Update complaint... | = M.   |         | UI     | To Do ▾ |

Name this sub-task

[Sub-task](#) ▾

↩

Cancel

Linked work items

Add linked work item

## 7.STORY 6

Jira work item

SCRUM25-2 / SCRUM25-7

1

## Mark bin as collected

To Do ⚡ ✦ Improve Story ... +

Description

As a Truck Driver,

I want to mark a bin as collected after completing the pickup,

So that the system accurately tracks completed waste collection.

Subtasks

0% Done

| Work                                                  | Pri... | Stor... | Ass... | Status  |
|-------------------------------------------------------|--------|---------|--------|---------|
| <a href="#">SCRUM25-21</a> Test route calculation...  | = M.   |         | U1     | To Do ▾ |
| <a href="#">SCRUM25-22</a> Verify route generation... | = M.   |         | U1     | To Do ▾ |

Linked work items

Add linked work item

Confluence content ⓘ

Product requirements

Try template

Details

Unassigned

## 8.Story 7

SCRUM25-3 / ☒ SCRUM25-9

1

Ensure fast route recalculation performance

To Do ▾

⚡ Improve Story

...

+

▼ Description

As a Sanitation Supervisor,  
  
I want route recalculation for 50 trucks and 10,000 bins to complete within 45 seconds,  
  
So that daily waste collection routes can be generated efficiently.

▼ Subtasks

⚡ Create suggested work items

Suggest

0% Done

| Work                                                                            | Pri... | Stor... | Ass...                   | Status  |
|---------------------------------------------------------------------------------|--------|---------|--------------------------|---------|
| <div><div></div><div>SCRUM25-23</div><div>Configure Sanitation...</div></div>   | =      | M.      | <div><div></div>Ui</div> | To Do ▾ |
| <div><div></div><div>SCRUM25-24</div><div>Configure Truck Driver...</div></div> | =      | M.      | <div><div></div>Ui</div> | To Do ▾ |

Name this sub-task

Sub-task ▾

↩

Cancel

Linked work items

Add linked work item

Confluence content

...

+

## 9.BACKLOG LIST ASSIGNMENT

|                                     |                                                       |              |       |   |    |               |
|-------------------------------------|-------------------------------------------------------|--------------|-------|---|----|---------------|
| <input type="checkbox"/>            | ▼ Backlog (7 work items)                              | 37           | 0     | 0 | 🔍  | Create sprint |
| <input checked="" type="checkbox"/> | SCRUM25-4 Generate optimized collection routes        | Smart Wa...  | To Do | 8 | ⬇️ | 👤             |
| <input checked="" type="checkbox"/> | SCRUM25-5 Manually modify collection routes           | Smart Wa...  | To Do | 5 | ⬇️ | 👤             |
| <input checked="" type="checkbox"/> | SCRUM25-6 View assigned collection route              | Waste Col... | To Do | 3 | ⬇️ | 👤             |
| <input checked="" type="checkbox"/> | SCRUM25-8 Manage missed-pickup complaints             | Complaint... | To Do | 5 | =  | 👤             |
| <input checked="" type="checkbox"/> | SCRUM25-10 Implement role-based system access         | Complaint... | To Do | 5 | ⬇️ | 👤             |
| <input checked="" type="checkbox"/> | SCRUM25-9 Ensure fast route recalculation performance | Complaint... | To Do | 8 | ⬇️ | 👤             |
| <input checked="" type="checkbox"/> | SCRUM25-7 Mark bin as collected                       | Waste Col... | To Do | 3 | =  | 👤             |
| + Create                            |                                                       |              |       |   |    |               |

## 10.BOARD

Spaces / Scrum\_banana\_problem#25\_

SCRUM25 board

Summary Timeline Active sprints Backlog Calendar List Forms Development Docs Reports

Search backlog

Version Epic

Quick filters

Clear filters

No epic

Smart Waste Collection Route Management

Waste Collection Execution and Tracking

Complaint Management and System Quality

+ Create epic

▼ SCRUM25 Sprint 2 5 Sep – 12 Sep (0 work items)

0 0 0 Start sprint

Plan your sprint  
Drag work items from the Backlog section or create new ones to plan the work for this sprint. Select Start sprint when you're ready.

+ Create

0 of 0 work items visible | Estimate: 0 of 0

▼ Backlog (4 work items)

10 8 0 Create sprint

SCRUM25-5 Manually modify collection routes

Smart Waste Colle...

In Progress

5

⬇️

👤

SCRUM25-6 View assigned collection route

Waste Collection E...

In Progress

3

⬇️

👤

SCRUM25-8 Manage missed-pickup complaints

Complaint Manage...

To Do

5

=

👤

SCRUM25-10 Implement role-based system access

Complaint Manage...

To Do

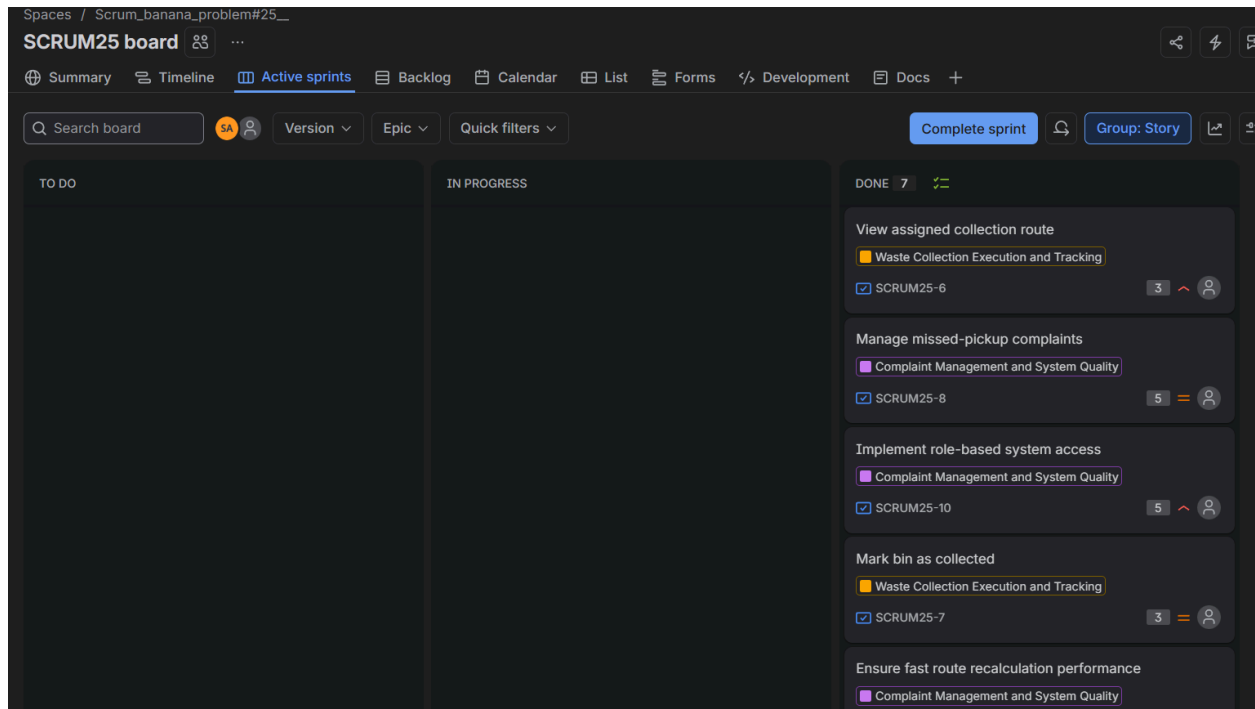
5

⬇️

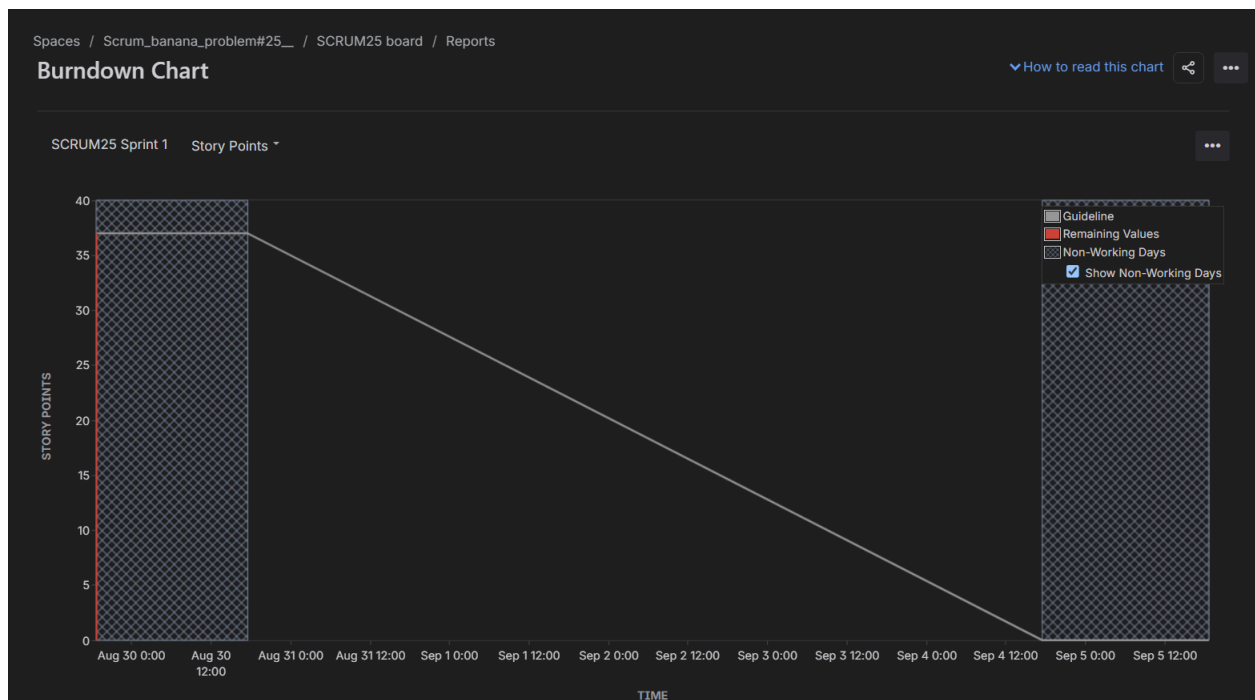
👤

+ Create

## 11. Active Sprint Board



## 11. Burndown Chart



## Reflection Questions:

1. Did your estimations reflect the actual effort?
2. Was your backlog well-prioritized?
3. How did your simulated sprint align with your plan?
4. What insights did the burndown chart give about your team's capacity?

## Answers:

1. The story point estimations generally reflected the relative complexity of the requirements. More complex tasks such as route optimization and performance optimization were assigned higher story points compared to simpler tasks such as viewing routes.
2. Yes. High-priority requirements related to route generation, collection tracking and system security were prioritized before medium-priority complaint management features.
3. The simulated sprint generally followed the planned workflow. User stories were moved from To Do to In Progress and finally to Done according to their planned priority.
4. The burndown chart showed how the remaining work decreased during the sprint. It provided an indication of whether the planned workload could be completed within the sprint duration and helped visualize the team's progress.